

High efficiency PAM4 ring modulator using W-Shaped PN junction

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ABSTRACT

A novel approach in designing the ring modulator's PN junction doping profile was proposed, where a W-shaped profile was used and compared to the conventional lateral profile, with the aim of achieving better performance and higher extinction ratios. Simulations were performed in the C-band on the 50GBd/s PAM4 ring modulator in the case of lateral doping profile and the case of the demonstrated approach of using a W-shaped doping profile. The results demonstrate how the implementation of the W-shaped profile introduces a significant 2.3x enhancement in the wavelength shift per unit voltage compared to its lateral counterpart, at the same doping levels, with lower losses and bandwidth. The demonstrated results show the effect of changing the doping profile on the performance of the ring modulator and its potential in making highly efficient optical modulators that can fulfil the growing market requirements. mod. eff.

Keywords: Silicon photonics, Ring modulator, PN junctions, Doping profiles

1. INTRODUCTION

Over the past decade, the demand for ultra-high-speed networks has grown significantly, leading to silicon photonics attracting substantial attention from the photonics industry, particularly in the fabrication of optical transceivers for data centers. For the case of high-speed requirements in datacenters, optical links are used to benefit from their high bandwidth. To use these optical links, the light needs to be modulated to carry the information required to be transferred.

Known for their compactness and ability to efficiently modulate light in photonic integrated circuits (PICs), ring modulators became a crucial component in silicon photonics. By doping the waveguide of the ring resonator, the round-trip parameters such as phase accumulation and transmission can be modulated through the plasma dispersion effect. Accordingly, the light circulating within the ring experiences varying phase accumulation depending on the applied voltage on the doped region (to be explained further in the upcoming section). In general, Ring modulator is more compact than other available choices like the Mach-Zehnder modulator and requires smaller driving voltage, making it compatible with CMOS technology.

The performance of the ring modulator, like any other, is highly affected by the doping profile implemented; the modulation efficiency can be enhanced by increasing the overlap and interaction between the optical mode and the free carriers. In this paper, a newly proposed W-shaped doping profile is implemented in the ring modulator and compared to its lateral counterpart.

2. THEORY AND DESIGN METHODOLOGY

To properly understand ring modulators, the main principle of the ring resonator and the basic concepts of modulation need to both be explained and discussed.

Ring Modulator Theory

The main principle of operation of a ring resonator is based on the interference of light within a ring cavity, where resonance in the cavity is achieved whenever the mode in the loop builds up a round-trip phase that equals an integer times 2π . The transmission spectrum of the bus waveguide with a single ring resonator exhibits dips around the ring resonances. Two basic configurations for the ring resonator are the all-pass configuration and the add-drop configuration as illustrated in Figure(1); in add-drop ring resonators, when the ring resonator is connected to two waveguides, a portion of the incident field is directed to the drop port making it advantageous because the second waveguide serves to "load" the resonator by adding extra loss. This approach balances the losses, enabling critical coupling and ensuring a high extinction ratio. The response of the add-drop ring resonator resembles a feedback mechanism where the light keeps on coupling from and to the ring, hence experiences a built-up round-trip phase that is an integer multiple of 2π and can be expressed as follows:

$$E_{thru} = \frac{\tau_1 - \tau_2 a e^{i\theta}}{1 - \tau_1 \tau_2 a e^{i\theta}} \quad (1)$$

$$E_{drop} = \frac{-\kappa_1 \kappa_2 \sqrt{a} e^{\frac{i\theta}{2}}}{1 - \tau_1 \tau_2 a e^{i\theta}} \quad (2)$$

Where κ, τ, a and θ are the cross-coupling coefficient, self-coupling coefficient, losses, and propagation phase respectively.

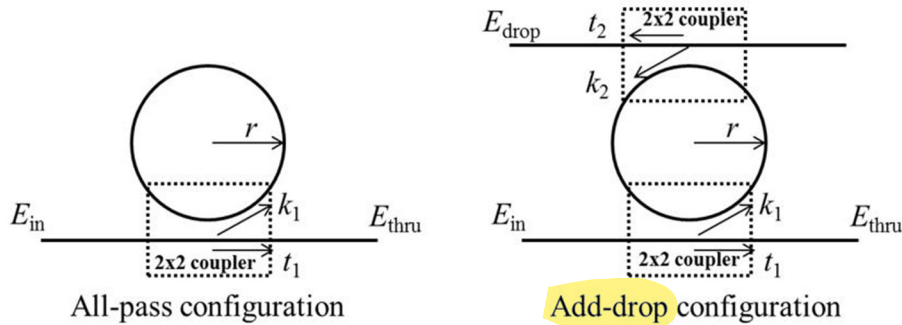


Figure 1: Ring resonator configurations

Most ring modulators operate by modulating intracavity parameters. In an intracavity ring modulator, the waveguide of the ring resonator (excluding the coupling region) is doped, enabling modulation of round-trip parameters such as phase accumulation and transmission through the plasma dispersion effect. As a result, the light circulating within the ring experiences varying phase accumulation and transmission depending on the voltage applied to the doped region. When the phase accumulation over a round trip reaches an integer multiple of 2π , the circulating light undergoes constructive interference with incoming light coupled from the bus waveguide, leading to light resonances within the cavity. Changes in the refractive index of the ring resonator waveguide consequently cause shifts in the resonance wavelength. This type of intensity modulation requires depletion of light from the ring resonator during the on-state and

injection of light into the resonator during the off-state. Consequently, the speed at which the device switches between these states depends on the response time of the cavity. When the quality factor (Q) of the cavity is high, the lifetime of light within the cavity is extended, resulting in longer rise and fall times for the field amplitude. Therefore, the operational bandwidth is constrained by the linewidth of the cavity. Reducing the photon lifetime and electrical time constant increases the bandwidth. However, a lower photon lifetime corresponds to a lower Q-factor, necessitating a higher driving voltage to achieve a desired Extinction Ratio (ER).

PN Junction Doping Profile

Lateral PN junctions have been dominating in optical modulation for a long time. They show high modulation speeds due to their reduced parasitic capacitance compared to other junction profiles. Lateral junctions have a limited depletion region, hence a limited overlap and interaction between the optical mode and the free carriers; although this widens the bandwidth due to the low capacitance, it causes lateral junctions to have relatively low modulation efficiency. The doping profile significantly influences the modulation efficiency; so, a novel doping profile is proposed and implemented in the ring modulator where the overlap between the mode and the depletion region is deliberately engineered to enhance the modulation efficiency while keeping the losses and junction capacitance within acceptable ranges. The newly proposed junction has a W-shaped profile as shown in Figure(2b), which compared to it lateral counter part illustrated in Figure(2a), has significantly larger overlap between the mode and the free carriers.

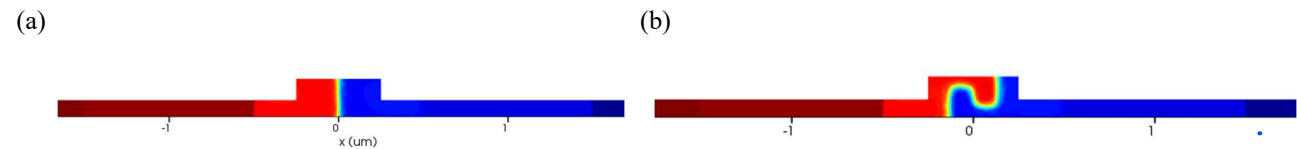


Figure 2: Rib waveguide cross section for (a) lateral PN junction and (b) proposed W-shaped PN junction

For proper junction characterization, the charge distribution in the junction is examined on finite element CHARGE and then implemented in the waveguide using finite difference FDE. Rigorous steady state CHARGE simulations were performed to optimize the junction doping concentrations and profile geometry, while also performing transient AC simulations to calculate the junction capacitance. The junction properties can be summarized in Table(1) and the function profile geometric parameters are clarified in Figure(3).

Table 1: W-shaped junction characteristics summary

	W-Shaped Junction	LPN junction
$N_e(cm^{-3}) = N_h(cm^{-3})$	5×10^{18}	
Interlaced width (nm)	125	-
Interlaced height (nm)	165	-
$V_{\pi} L$ (V.cm) at $V_{bias} = -2V$	0.24	0.59
Losses (dB/mm) at $V_{bias} = -2V$	12.6	15.2
Capacitance (f/cm) at $V_{bias} = -2V$	1.6	0.7

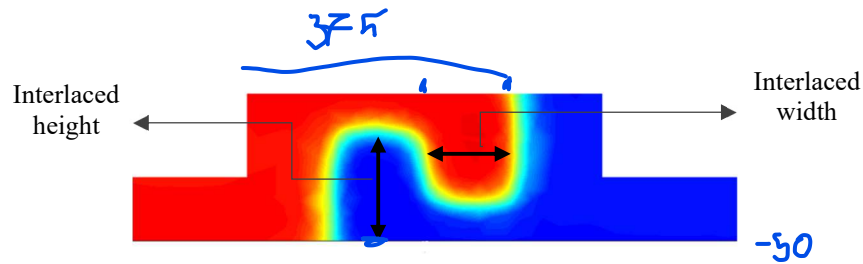


Figure 3: W-shaped junction cross-section

The final W-shaped junction design has longer junction, thus its capacitance increases where it shows a 2.3x enhancement in modulation efficiency due to greater optical mode overlap with junction. It also exhibits lower losses with a degradation in RC bandwidth at a bias voltage of -2V compared to the lateral junction for the same doping concentrations.

Ring Modulator Design

The Design flow consists of firstly specifying the design parameters, then performing the required simulations on Ansys Lumerical. Starting by CHARGE simulations to study the PN junction from the doping density and profile to the junction resistance and capacitance, then implementing the chosen charge profile in the waveguide of choice on FDE to get the variation of the effective refractive index with the applied voltage. Afterwards, and based on the chosen design specs and parameters, the coupling regions of the ring modulator are designed accordingly. For the final stage, the ring modulator is implemented in MATLAB to perform frequency domain system simulations hence calculate the extinction ratio, insertion loss and electro-optical bandwidth.

After discussing the main principle of the ring modulator operation, it can be concluded that the design parameters are the coupling coefficients of the ring, its radius, the doping density and profile of the PN junction, and of course the geometry of the waveguide itself. Firstly, the waveguide is chosen for a 10um radius ring to have typical dimensions of a 220nm core height, 500nm core width, and 90nm slab thickness. Then, the doping level is chosen around $5 \times 10^{18} \text{ cm}^{-3}$; such high doping levels are to compensate for the relatively short optical path of the ring to achieve high modulation efficiency.

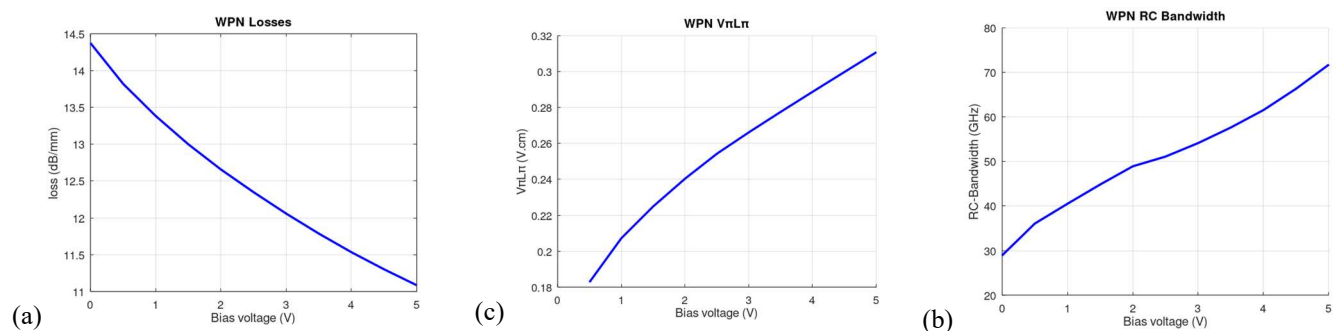


Figure 4: WPN-junction, (a) losses, (b) $V\pi L\pi$ and (c) RC-bandwidth

MRM design parameters are coupling and detune from resonance. Where for WPN junction these parameters were swept and ER and OMA at input power of 7 dBm were reported on a contour map in Figure (5).

Optical Modulation Amplitude, quantifies the difference in optical power between the logical "1" and "0" levels (in case of PAM4, between adjacent signal levels).

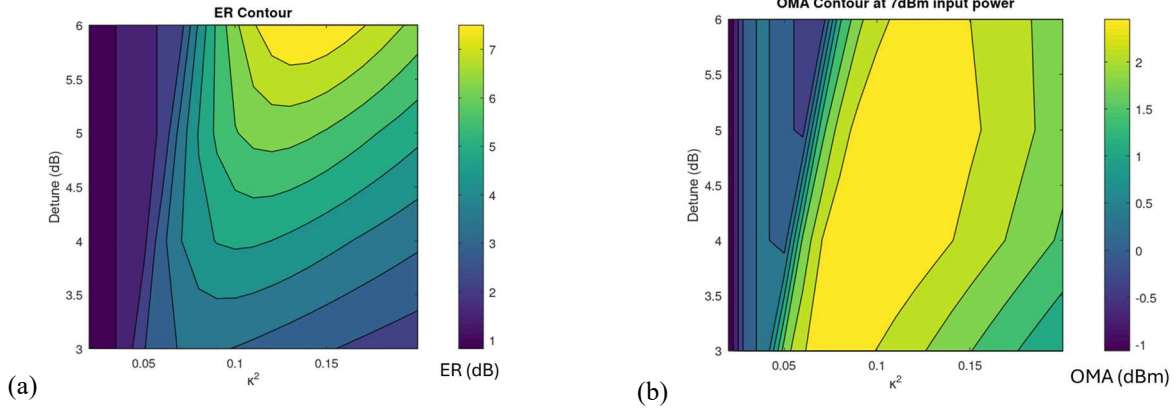


Figure 5: contour maps for MRM, (a) ER, (b) OMA

A quality factor of about 2600 is chosen for a target of 50GBd/s, where the quality factor can be calculated as:

$$Q = \frac{n_g 2\pi^2 R \sqrt{\tau a}}{\lambda(1-\tau)} \quad (3)$$

From the previous equation, the self-coupling coefficient τ can be easily extracted for the selected quality factor as illustrated in Figure (6). Accordingly, κ is chosen to be 17% which corresponds to a ring-to-waveguide gap of 175nm.

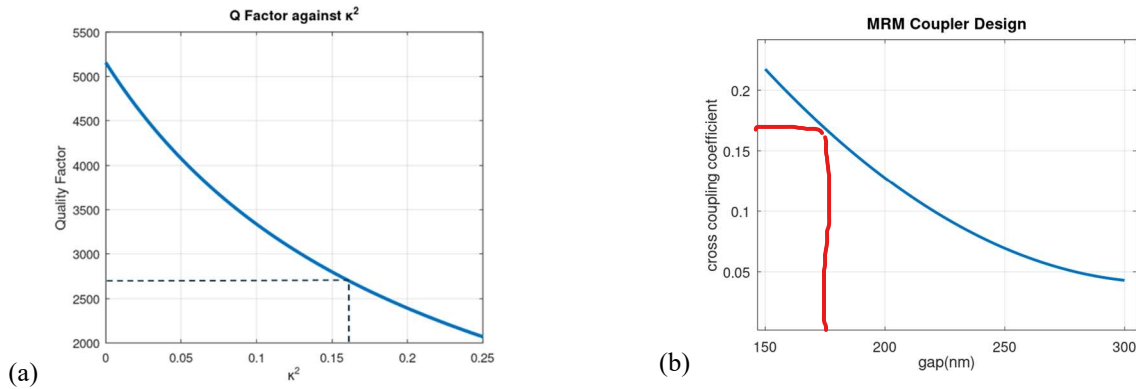


Figure 6: (a) Quality factor against transmission coefficient (b) Transmission vs. gap distance

There are three operating conditions for the resonator. The first condition is the critically coupled equation which occurs when the following condition occurs: $\tau_1 = a\tau_2$. This equation means that this condition takes place when external loss due to the coupling in the through port (τ_1) is equal to the cavity internal loss ($a\tau_2$). The other two conditions are the over coupled condition and the under coupled condition. The under-coupled regime takes place when $\tau_1 > a\tau_2$, while the over-coupled regime takes place when $\tau_1 < a\tau_2$. The extinction ratio is highest at the critically coupled condition hence it is chosen as the operating condition for the ring.

3. RESULTS AND DISCUSSION

Based on the theory and design methodology discussed, the design of the ring modulator chosen in the previous section, along with both the lateral junction and the novel W-shaped junction profile, are modeled in frequency domain using the setups stated. A PAM4 modulation scheme is chosen for the ring modulator setup.

The transmission spectrum against wavelength can be plotted where the wavelength shift per unit voltage is calculated and compared for both junctions as illustrated in Figure (7).

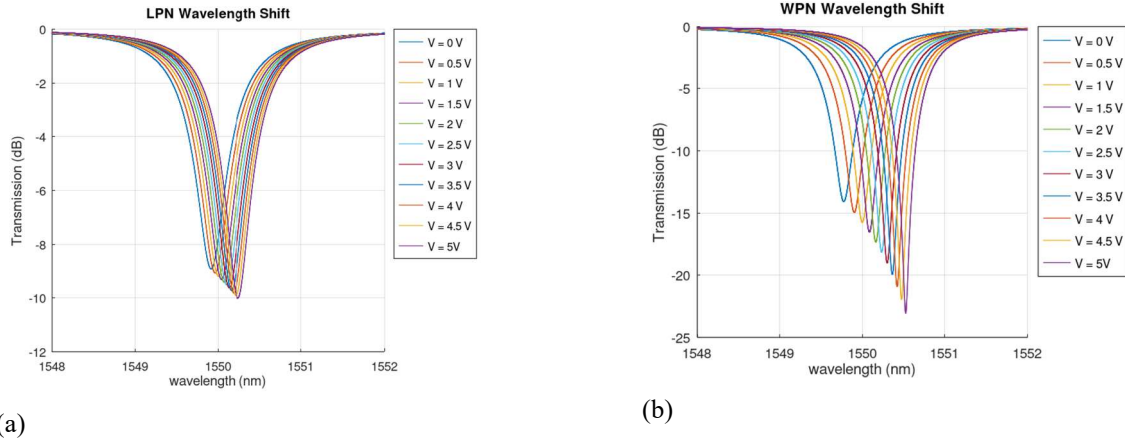


Figure 7: Transmission spectrum against wavelength for different bias voltage for (a) lateral junction and (b) W-shaped junction

As explained previously, increasing the overlap between the depletion region and the mode is expected to enhance the modulation efficiency hence enhancing the wavelength shift per unit voltage as shown in the results illustrated in Figure (5). The ring modulator using the lateral shaped junction exhibits 70pm/V wavelength shift while the newly proposed W-shaped junction shows a significant enhancement with a wavelength shift of 163pm/V at the doping concentrations.

The design of the MRM was optimized, to get best ER, OMA and BW possible. The MRM was detuned to be at -5.5 dB IL, biased at 1 volt, achieving ER of 6.57 dB, OMA of 2.3 dBm and BW 38 GHz.

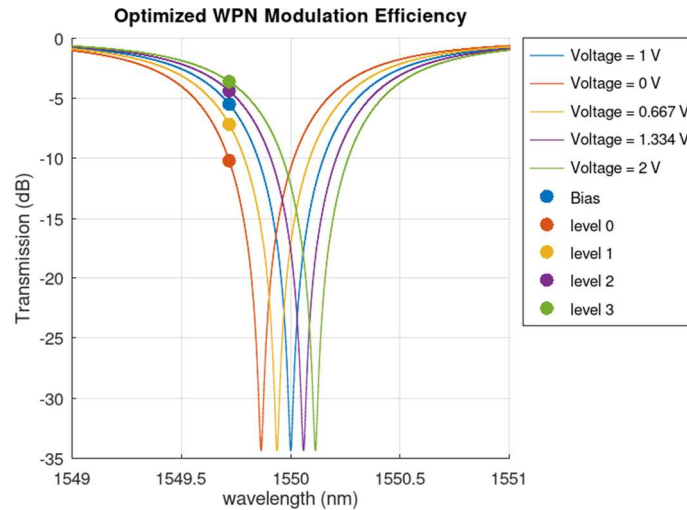


Figure 8: Transmission spectrum against wavelength for the optimized design

Table 2: summary of results.

ER (dB)	OMA (dBm)	BW (GHz)
6.57	2.3	38

4. CONCLUSION

A novel approach in designing the ring modulator's PN junction doping profile was proposed, where a W-shaped profile was used and compared to the conventional lateral profile, with the aim of achieving better performance and higher extinction ratios. Simulations were performed on the 50GBd/s PAM4 ring modulator in the case of lateral doping profile and the case of the demonstrated approach of using a W-shaped doping profile. The results demonstrate how the implementation of the W-shaped profile introduces a significant enhancement in the wavelength shift per unit voltage and modulation efficiency and lower losses compared to its lateral counterpart, at the same doping levels, with slight degradation in bandwidth. This approach shows to be promising in the performance improvement of optical modulators.

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